Journal of Human Nutrition and Dietetics (1996), **9**, 303–307

Does provision of additional dietary information affect actual or only reported compliance to a low-fat diet over 12 weeks in hyperlipidaemic individuals? Report of a pilot study

Clare Grace and Carolyn Summerbell*
*Dietetic Department, The Royal Hospitals Trust, and *Rank Department of Human Nutrition, St Bartholomew's and Royal London Hospital School of Medicine and Dentistry, London, UK*

This randomized controlled pilot study was designed to measure the effect of additional dietary information on change in reported percentage energy from fat (% fat), total daily energy intake (TDEI), body mass index (BMI) and serum cholesterol in 13 hyperlipidaemic subjects, referred from the Cardiology Unit of St. Bartholomew's Hospital, over 12 weeks. At baseline (visit 1), % fat and TDEI were assessed using a validated food frequency questionnaire (Paisley *et al.*, 1996), and BMI and fasting lipids were measured. 'Standard' low-fat dietary advice was given to all patients. Subjects were randomized at 4 weeks (visit 2) to receive either further 'standard' dietary advice (control group) or further 'standard' dietary advice plus an additional information package (intervention group). At 12 weeks (visit 3), % fat, TDEI, BMI and fasting lipids were reassessed.

Both groups reported a significant reduction in % fat; a 23% reduction ($P=0.00$) in the intervention group, and an 11% reduction ($P=0.005$) in the control group. However, there was no significant difference in the reported % fat at the end of the study between the two groups ($P=0.57$). Plasma cholesterol was not significantly reduced in either the intervention or the control group (mean 7.3–6.7 mmol/l, 7.2–6.6 mmol/l, respectively). Reported energy intake was significantly reduced in both groups, but the extent of energy reduction was not reflected in the degree of weight lost. These results suggest that the provision of additional dietary information affects reported rather than actual compliance to a low-fat diet.

Key words: hypercholesterolaemia, low-fat diet, dietary compliance, dietary advice, serum cholesterol.

Introduction

Metabolic studies have confirmed the effectiveness of a low total and saturated fat diet on plasma cholesterol reduction (Cobb & Teitlebaum, 1994). However, the use of dietary manipulation as the first line of treatment for hyperlipidaemia has recently been criticized due to the proposed ineffectiveness of present dietary restrictions (Ramsay *et al.*, 1991). Clearly the efficacy of a low-fat diet largely depends on compliance. It has been suggested that compliance to a low-fat diet may be improved by extending the amount and the range of dietary information provided (McCann, 1990).

Correspondence: Clare Grace, Dietetic Department, The Royal Hospitals Trust, Whitechapel, London E1 1BB, UK.

The aim of this randomized controlled pilot study was to test whether additional dietary information improved actual, or only reported, compliance to a low-fat diet in hyperlipidaemic individuals who were not already consuming a diet containing less than 30% of energy from fat. Previous studies have observed under-reporting of energy intake (Bingham, 1987; Heitmann & Lissner, 1995; Summerbell *et al*., 1995). Therefore, the biomarkers used in this study (change in body weight and change in total plasma cholesterol, TC) were used to assess the validity of reported change in total daily energy intake (TDEI) and percentage energy from fat (% fat), respectively. Reported change in % fat was compared to change in TC, while controlling for changes in body weight, because body weight and TC are independently correlated; a 10% decrease in body weight can be taken as incurring a 15% reduction in TC (Anderson *et al*., 1995).

Methodology

Ten male and 4 female caucasian hyperlipidaemic subjects, aged 35–65 years with a body mass index (BMI = weight (kg)/ height (m)^2) of between 25 and 35 were recruited from patients referred by the cardiac rehabilitation unit. Patients were recruited regardless of the type of hyperlipidaemia or whether the hyperlipidaemia was familial. All subjects met the following criteria; a reported baseline diet of more than 30% of energy from fat, no significant other medical conditions known to independently affect lipid profile, following no other therapeutic diet or taking hypolipidaemic medication. One male subject (who had been randomized to the control group) was retrospectively removed from the study due to a reported baseline diet of less than 30% energy from fat.

At baseline BMI, fasting lipid profile, TDEI and % fat were assessed. TC and total plasma triglyceride (TG) levels were measured using the enzymatic colorimetric test involving the oxidase peroxidase reaction (Trinder *et al*., 1969). The high-density lipoprotein (HDL) fraction was determined by precipitation of chylomicrons followed by centrifugation of the supernatent (Study Group of the European Atherosclerosis Society, 1987) and the low-density lipoprotein (LDL) fraction was calculated using the Friedewald equation (Friedewald, 1972).

TDEI and total daily fat intake (TDFI) were assessed using a validated food frequency questionnaire (FFQ) (Paisley *et al*., 1996). This FFQ was used to assess TDEI and TDFI rather than a standard 7-day weighed record because we wanted to capture intake data over a longer period of time than just 7 days. ‘Standard’ low-fat dietary advice was given to all patients (Thomas, 1994).

At visit 2 (4 weeks from baseline) subjects with a baseline TC level greater than 6.5 mmol/l who also reported a diet containing more than 30% energy from fat were randomized to receive either further ‘standard’ dietary advice (control group) or further ‘standard’ dietary advice plus an additional information package (treatment group). The additional information package contained details on improving the practical implementation of a low-fat diet, such as low-fat cooking methods, low-fat recipe adaptation and eating out on a low-fat diet. The package was self-explanatory, required minimal verbal explanation and did not extend interview time by more than 10–15 min. Subjects were separated according to gender, and within each gender block randomization using blocks of six was used to allocate individuals to the two groups. At visit 3 (12 weeks from baseline) BMI, fasting lipids, TDEI and % fat were assessed.

The degree of under-reporting of energy intake at baseline was calculated by comparing measured TDEI with predicted TDEI; predicted TDEI was calculated using the Schofield equation (Schofield *et al*., 1985) and assuming a physical activity level of 1.4 (Goldberg *et al*., 1991).

Expected weight loss over the 12-week study period was compared with observed weight loss. Expected weight loss was estimated by first calculating the deficit in TDEI over the entire 12 weeks, which was defined as predicted TDEI at baseline (see above) minus reported TDEI at 12 weeks multiplied by 84 (number of days in 12 weeks). Expected weight loss was defined as the deficit in TDEI over 12 divided by 7000 (because 7000 kcals = 1 kg weight loss; Garrow, 1988).

Table 1. Change in reported fat and energy intake, body weight and lipid profiles in hyperlipidaemic subjects over a 12-week period

	Change % fat	Change REI (kJ)	%UR (baseline)	Weight change (kg)	Change TC (mmol/l)	Change LDL (mmol/l)	Change HDL (mmol/l)	Change TG (mmol/l)
Intervention Group ($n = 8$)								
M1	27	−2831	−16	−2	−1.3	−1.2	−0.2	0
M2	22	−3839	5	1	−1.1	−0.7	−0.2	−0.3
M3	23	−4792	38	−3	NA	NA	NA	NA
M4	16	−4393	27	−2	0.1	0.1	−0.5	1.1
M5	35	−5342	−8	−3	−1.6	−1.7	0.4	−0.4
M6	27	−6661	−5	−4	0.4	0.5	0.2	−0.4
F1	24	−2717	−4	0	0.5	1.2	0	−1.4
F2	17	500	−27	−10	−1.3	−0.9	−0.7	0.6
Mean (SD)	24 (6)	−3759 (2155)	1 (21)	−2.9 (3.3)	−0.6 (−0.9)	−0.4 (1)	−0.2 (0.4)	−0.1 (0.8)
Control group ($n = 5$)								
M7	8	−5174	17	1	0.5	NA	0.3	0.4
M8	12	−2432	−4	−2	−0.2	−0.4	0.2	−0.1
M9	9	−2503	−10	−2	−0.8	−0.9	0.3	−0.3
F3	10	−2125	−20	−3	−1.4	−1.2	−0.1	−0.2
F4	20	−2767	14	−2	−1.1	−1	−0.1	−0.6
Mean (SD)	12 (5)	−2995 (1239)	−1 (16)	−1.6 (1.5)	−0.6 (0.8)	−0.9 (0.3)	−0.2 (0.4)	−0.1 (0.3)

M = male; F = female; % fat = reported percentage energy from fat; NA = not available; TC = total cholesterol; LDL = low-density lipoprotein; HDL = high-density lipoprotein; TG = triglycerides; REI = reported energy intake; UR = % under-reporting of energy intake.

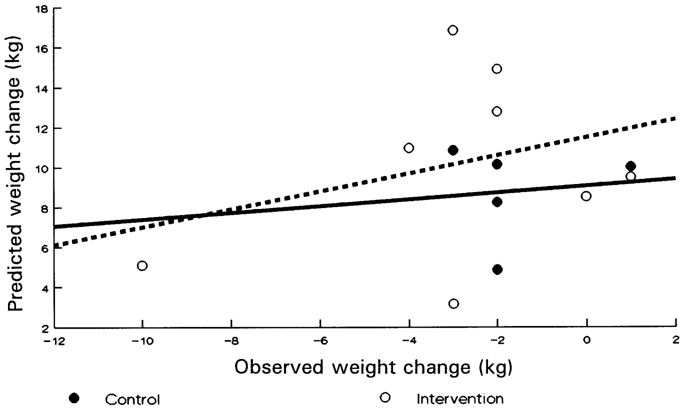

Fig. 1. Predicted compared with observed weight loss over 12 weeks.

Methods of analysis

Differences within the groups over time were assessed using the paired *t*-test. Differences between groups at one point in time were assessed using the unpaired *t*-test.

Results

Of the 14 subjects recruited to the trial, only one was retrospectively removed due to low reported baseline % fat (defined as less than 30% energy from fat). Of the remaining 13 subjects, all completed the trial. The results are shown in Table 1. The unequal distribution of male and female subjects in the intervention and control groups was a result of using block randomization in batches of six for each gender separately; in retrospect it may have been better to use smaller blocks for randomization.

The intervention group reported a significantly higher ($P = 0.045$) baseline % fat (mean 50%, SD 7) than the control group (mean 42%, SD 3). Both groups reported a significant reduction in % fat over the study period; 24% reduction ($P = 0.000$) in the intervention group, and 12% reduction ($P = 0.005$) in the control group. The reduction in % fat was greater in the intervention group compared with the control group ($P = 0.003$), but there was no difference ($P = 0.57$) in the reported % fat at the end of the study between the two groups.

Plasma cholesterol was not significantly reduced in either the intervention group (7.3 ± 0.3 mmol/l, mean $\pm$ SD to 6.7 ± 0.7 mmol/l, $n = 7$, one result missing) or the control group (7.2 ± 0.9 mmol/l to 6.6 ± 0.6 mmol/l, $n = 5$). There was no correlation between the reported reduction in % fat and the actual reduction in TC in either group, even though the intervention group reported a two-fold (24%) reduction in TDFI compared with the control group (12%).

LDL was not significantly reduced in either the intervention group (5.0 ± 0.5 mmol/l to 4.6 ± 0.6 mmol/l, $n = 7$, one result missing) or the control group (4.9 ± 1.0 mmol/l to 4.5 ± 0.5 mmol/l, $n = 4$, one result missing). HDL was not significantly reduced in either the intervention group (1.3 ± 0.5 mmol to 1.1 ± 0.2 mmol/l, $n = 7$, one result missing) or the control group (1.2 ± 0.4 mmol/l to 1.3 ± 0.4 mmol/l). TG was not significantly reduced in either the intervention group (2.3 ± 1.2 mmol/l to 2.2 ± 0.9 mmol/l, $n = 7$, one result missing) or the control group (2.5 ± 1.2 mmol/l to 2.3 ± 1.4 mmol/l).

The degree of under-reporting of energy intake at baseline was extremely variable, but not biased in one direction (range −27% to 38%) or associated with baseline BMI. Furthermore, there appeared to be no difference in either the degree or direction of under-reporting of energy intake between the intervention and control groups.

Reported TDEI was reduced in both the intervention (9941 ± 2528 kJ/day to 6182 ± 1193 kJ/day, $P = 0.002$) and the control group (9261 ± 1848 kJ/day to 6262 ± 1172 kJ/day, $P = 0.006$) over the 12-week period, but the extent of calorie reduction was not reflected by the degree of weight loss (intervention group 75 ± 10 kg to 72 ± 9 kg, no significant; control group 72 ± 9 kg to 70 ± 9 kg, not significant). Furthermore, there appeared to be no correlation between expected and observed weight loss in either group (Fig. 1).

Interpretation of these results proves difficult for several reasons. Firstly, the well documented potential for large subject error in the reporting of food intake (Heitmann & Lissner, 1995; Schoeller, 1990) may have had a significant effect on these results. Secondly, individual variation in LDL responsiveness to a low-fat diet can be considered as a further complication (Denke, 1995). Lack of correlation between change in TDEI and change in body weight suggests a reporting error rather than individual variation in LDL responsiveness to changes in fat intake.

Conclusion

These results suggest that the provision of additional dietary information affects reported rather than actual compliance to a low-fat diet. Previous studies have observed selective under-reporting of total fat and carbohydrate intake, particularly amongst the obese (Heitmann & Lissner, 1995). These specific errors in dietary reporting may be due to under-estimation of foods considered unhealthy. Additional emphasis in the intervention group on the reduction of fatty foods could have resulted in a greater under-reporting of fat intake.

References

Anderson, R.E. Wadden, T.A., Bartlett, S.J., Vogt, R.A. & Weinstock, R.S. (1995) Relation of weight loss to changes in serum lipids and lipoproteins in obese women. *Am. J. Clin. Nutr.*. **62**, 350–357.

Bingham, S.A. (1987) The dietary assessment of individuals; methods, accuracy, new techniques and recommendations. *Nutr. Abs. Rev.* (Series A) **57**(10), 705–742.

Cobb, M.M. & Teitlebaum, H. (1994) Determinants of plasma cholesterol responsiveness to diet. *Br. J. Nutr.* **71**, 271–282.

Denke, M.A. (1995) Review of human studies evaluating individual dietary responsiveness in patients with hypercholesterolaemia. *Am. J. Clin. Nutr.* **62**, 471–477.

Friedewald, W.T. (1972) Estimation of plasma low density lipoprotein cholesterol concentration without the use of the preparative ultracentrifuge. *Clin. Chem.* **18**, 499–500.

Garrow, J.S. (1988) *Obesity and related diseases*. London: Churchill Livingstone.

Goldberg, G.R., Black, A.E., Jebb, S.A., Cole, T.L., Murgatroyd, P.R., Coward, W.A. & Prentice, A.M. (1991) Critical evaluation of energy intake using fundamental principles of energy physiology: 1. Derivation of cut-off limits to identify under-recording. *Eur. J. Clin. Nutr.* **45**, 569–581.

Heitmann, B.L. & Lissner, L. (1995) Dietary under reporting by obese individuals—is it specific or non-specific? *Br. Med. J.* **311**, 986–989.

McCann, B.S., Retzlaff, M.P.H., Dowdy, A.A., Waldon, C.E. & Knopp, R.H. (1990) Promoting adherence to low fat, low cholesterol diets: review and recommendations. *J. Am. Dietet. Assoc.* **90**(10), 1408–1417.

Paisley, C.M., Lloyd, H.M., Brown, W. & Mela, D.J. (1996) Production and validation of a food frequency questionnaire to estimate dietary fat intake. *J. Hum. Nutr. Dietet.* **9**, 189–206.

Ramsay, L.E., Yeo, W.W. & Jackson P.R. (1991) Dietary reduction of serum cholesterol concentration; time to think again. *Br. Med. J.* **303**, 953–957.

Schoeller, D.A. (1990) How accurate is self-reported dietary energy intake? *Nutr. Rev.* **48**, 373–379.

Schofield, W.N., Schofield, C. & James, W.P.T. (1985) Basal metabolic rate. *Hum. Nutr. Clin. Nutr.* **39C** (Suppl. 1), 1–96.

Study Group of the European Atherosclerosis Society (1987) Strategies for the prevention of coronary heart disease: a policy statement of the European Atherosclerosis Society. *Eur. Heart J.* **8**, 77–88.

Summerbell, C.D., Moody, R.C., Shanks, J., Stock, M.J. & Geissler, C. (1995) Sources of energy from meals versus snacks in 220 people in four age groups. *Eur. J. Clin. Nutr.* **49**, 33–41.

Trinder, P. (1969) Enzymatic hydrolysis of triglycerides with subsequent determination of liberated glycerol by colorimetry. *Ann. Clin. Biochem.* **6**, 24.

Thomas, B., ed. (1994) Hyperlipidaemia. In *Manual of Dietetic Practice*. pp. 468–474. Oxford: Blackwell Scientific Publications.

Submitted December 1995, *Revised* February 1996, *Accepted* April 1996.